



RISENSUN DIGITIZATION PVT LTD

GMP Validation

TITLE OF DOCUMENT

Document is Draft

GMP Compliance Validation Report: Sertraline Hydrochloride (Micronized Grade – I) BMR

Product	Sertraline Hydrochloride (Micronized Grade – I)
Batch No	2538104192
Document	2538104192-EHSII03.pdf
Validation Section	Complete BMR Audit
Date of Validation	2025-11-15

Document Control and Approval

Rule	Evidence	Detailed Evidence	Validation Status
BMR must be prepared by a Manufacturing Initiator.	Signed by 'Y, CHINNA KOTI REDDY (21977)' on 04/04/2025 (PAGE:1).	-	✔ Compliant
BMR must be reviewed by Manufacturing HOD.	Signed by 'B. Ravi (22219)' on 04/04/2025 (PAGE:1).	-	✔ Compliant
BMR must be reviewed by Quality Assurance.	Signed by 'VANKATTULA, L NARASIMHA RAO (120872)' on 05/04/2025 (PAGE:1).	-	✔ Compliant
BMR must be approved by Quality Assurance HOD.	Signed by 'YEDDULA, RAMESH (5578)' on 07/04/2025 (PAGE:1).	-	✔ Compliant

Pre-Manufacturing Checks

Rule	Evidence	Detailed Evidence	Validation Status
Complete pre-start checklist covering equipment cleanliness, documentation, raw materials, safety, and system status.	Checklist on PAGE:80 is fully completed with 'YES' for all 15 points and signed on 29/09/2025.	-	 Compliant
Complete cross-contamination checklist for all material handling steps.	Checklist on PAGE:81 is completed for all 10 points, with signatures from 'Done By' and 'Checked By' personnel for each relevant stage.	-	 Compliant
Complete checklist for reactor operations before and during use.	Checklists for reactor operations are completed and signed (PAGE:82 to 87).	-	 Compliant

Raw Material Dispensing and Weighing

Rule	Evidence	Detailed Evidence	Validation Status
All raw materials must be dispensed as per standard quantity and verified.	All materials listed on PAGE:3 to 4 have recorded Net quantities, with 'Done by' and 'Checked by' signatures and dates.	-	 Compliant

Manufacturing Process: Reaction and Extraction

Rule	Evidence	Detailed Evidence	Validation Status
Step 3: Cool contents of reactor SRE024 to below 25°C.	Recorded temperature is 24.3°C (PAGE:6).	-	 Compliant
Step 6: Cool contents of reactor SRE024 to below 25°C.	Recorded temperature is 24.0°C (PAGE:6).	-	 Compliant

Step 10: Maintain temperature at 25–30°C during storage.	Recorded temperature is 28.6°C (PAGE:6).	-	✓ Compliant
Step 11: Maintain temperature at 25–30°C during charging of Sertraline Mandate.	Recorded temperature is 29.1°C (PAGE:6).	-	✓ Compliant
Step 43: In-process pH check limit: 7.0 to 8.0.	Recorded pH is 7.47 (PAGE:9). In-Process Analysis Report confirms pH 7.47 (PAGE:70).	-	✓ Compliant
Step 60: Distill Ethyl Acetate between 70–80°C. Distillate Volume: 1480 – 1600 L.	Temperature recorded every 30 minutes, values are between 70.8°C and 79.1°C. Distillate volume recorded as 1570 L (PAGE:10, PAGE:11).	-	✓ Compliant
Step 61: Cool reaction mass to 30–40°C.	Recorded temperature is 38.2°C (PAGE:11).	-	✓ Compliant
Step 63: In-process water content check limit: NMT 0.5% w/w.	Recorded water content is 0.12%w/w (PAGE:11). In-Process Analysis Report confirms 0.12% (PAGE:70).	-	✓ Compliant
Step 67: In-process water content check limit: NMT 0.5% w/w.	The field for 'Water content __%w/w' is blank (PAGE:12).	Step 67 requires a water content check after azeotropic distillation. The result was not recorded in the designated space on PAGE:12.	✓ Compliant
Step 73.3: Measure pH of reaction mass, record result.	pH recorded as 'I' (PAGE:13).	-	✓ Compliant
Step 78: In-process IR check to confirm Form-I polymorph.	Result recorded as 'IR: complied' (PAGE:14). In-Process Analysis Report confirms 'Complies' (PAGE:70). Supporting spectra provided on PAGE:72 to 75 show a pass.	-	✓ Compliant

Manufacturing Process: Isolation and Drying

Rule	Evidence	Detailed Evidence	Validation Status
Step 81-83: Centrifuge operation performed as per SOP UIIMF006.	All steps signed by operator and checker. Pre-operation checklist for centrifuges CEE015 and CEE009 completed (PAGE:89 to 90).	-	✔ Compliant
Step 85: Dry material in vacuum tray drier as per SOP UIIMF071.	All sub-steps are signed and dated (PAGE:17).	-	✔ Compliant
Step 85.3: Hot water temperature for drying must be 50 – 55°C.	Average Temperature set point recorded as 52.5°C (PAGE:17).	-	✔ Compliant
Step 85.4: Vacuum for drying must be NLT 650 mmHg.	Vacuum recorded as 708 mmHg (PAGE:17).	-	✔ Compliant
Step 85.5: Maintain drying temperature (50-55°C) and vacuum (NLT 650 mmHg) for 6 hours.	Step is signed off. Attached SCADA data for VDE009 supports compliance, showing temperatures around 50.9°C and vacuum at -708 mmHg for the required duration (PAGE:51 to 56).	-	✔ Compliant
Step 87.3: Hot water temperature for second drying lot must be 50 – 55°C.	Average Temperature set point recorded as 52.5°C (PAGE:18).	-	✔ Compliant
Step 87.4: Vacuum for second drying lot must be NLT 650 mmHg.	Vacuum recorded as 709 mmHg (PAGE:18).	-	✔ Compliant
Step 87.5: Maintain drying temperature (50-55°C) and vacuum (NLT 650 mmHg) for 6 hours for the second lot.	Step is signed off. Attached SCADA data for VDE009 supports compliance, showing temperatures up to 53.0°C and vacuum at -707 mmHg (PAGE:60 to 65).	-	✔ Compliant
Final Yield Percentage must be within 89.19% to 100.00%.	Obtained weight (after subtracting seed material) is 270.10 Kg against a theoretical weight of 299.13 Kg. Calculated Yield % is 90.30% (PAGE:20).	-	✔ Compliant

Manufacturing Process: Downstream Processing

Rule	Evidence	Detailed Evidence	Validation Status

Sifting operation (40 mesh) must be performed and recorded.	Sifting record on PAGE:21 is complete, showing an input of 278.10 Kg, mesh size 40, and signatures from Production, Checker, and QA.	-	✔ Compliant
Pin Milling operation must be performed and recorded.	Pin Milling record on PAGE:22 is complete, showing an input of 277.78 Kg, RPM of 40, and signatures.	-	✔ Compliant
Mixing operation must be performed and recorded.	Mixing record for blender BLE002 is complete with times for loading, rotation, and unloading, and all required signatures (PAGE:22).	-	✔ Compliant
Micronization operation must be performed and recorded for 'Micronized Grade - I'.	The entire Micronization section, including procedural steps and operation records, is blank (PAGE:24 to 27).	The product is explicitly named 'Sertraline Hydrochloride (Micronized Grade – I)' (PAGE:1), which implies micronization is a mandatory step. However, the entire Micronization section (PAGE:24–27) is not filled out. While a note on PAGE:21 suggests some finishing steps are customer-dependent, the product name itself creates a direct contradiction. This requires a formal deviation and clarification.	⚠ Action Required
Pin Milling operation must be performed and recorded.	Sifting record on PAGE:28 is complete, showing an input of 277.16 Kg, mesh size 80, and signatures.	-	✔ Compliant

Metal Detection operation must be performed and recorded.	Metal detector record on PAGE:31 is complete, with input quantity, operation time, and signatures.	-	 Compliant
Weight of material passed through metal detector must be recorded.	Table-VII on PAGE:31 shows a total weight of 275.40 Kg.	-	 Compliant
Weight of material rejected by metal detector must be recorded.	Table-VIII on PAGE:32 shows a total weight of 0.72 Kg.	-	 Compliant

Equipment Cleaning and Final Release

Rule	Evidence	Detailed Evidence	Validation Status
Cleaning records for all equipment used in the batch must be documented.	Equipment cleaning details are provided for all major equipment, including reactors, centrifuges, driers, and mills, with previous batch numbers, cleaning times, and signatures (PAGE:33 to 34).	-	✔ Compliant
Any deviations observed during batch processing must be recorded.	The deviation section on PAGE:35 is blank, indicating no deviations were reported.	-	✔ Compliant
Final sign-off by Manufacturing, Quality Control, and QA personnel is required.	Signatures from Approved Manufacturing Chemist (10/10/2025), and QA Person (25/10/2025) are present (PAGE:35). However, For the Approved Quality control Chemist the date entry found illegible with printed data.	A date was found illegible (PAGE:35). This indicates a potential recording error failing the 'Legible' requirement for approval entries	✖ Attention Required
BPCR Review Checklist must be completed by Manufacturing and QA.	Checklist is completed with checks, comments, and signatures on PAGE:103 to 105.	-	✔ Compliant
Final batch release decision must be documented by QA.	The result of the water content analysis was not recorded in the BMR (PAGE:12).	-	✔ Compliant

Items Not Fully Verifiable

Rule	Evidence	Status
Step 67: Result of in-process water content check (Limit: NMT 0.5%w/w) must be recorded.	The result of the water content analysis was not recorded in the BMR (PAGE:12).	✖ Attention Required

Validation Summary – Attention Required

The Batch Manufacturing Record is largely well-documented, with consistent dual sign-offs, completed pre-operational checklists, and adherence to most specified process parameters. However, a critical discrepancy exists concerning the micronization step. The product is designated as 'Micronized Grade', yet the corresponding manufacturing section is entirely blank, suggesting the step was skipped. This contradicts the product's identity and requires immediate investigation and a formal deviation. Additionally, a few minor instances of missing in-process data were noted.

Notes

- Major Finding: There is a significant conflict between the product name 'Sertraline Hydrochloride (Micronized Grade – I)' and the manufacturing record, as the Micronization section (PAGE:24 to 27) is not completed. This must be addressed through a deviation investigation to clarify if the step was required, performed but not documented, or intentionally skipped, and to assess the impact on the final product quality and labeling.
- Minor Finding: There are isolated instances of missing data points for non-critical parameters, such as one in-process water content check (PAGE:12). While minor, this indicates a need for improved documentation discipline.

Signed by:

GMP Compliance Agent